

06 — Cost Calculation & Assumptions

Per the requirement, this folder contains (a) the **assumptions** behind the estimate and (b) a **cost calculation**. The official figures should be produced with the AWS Pricing Calculator at <https://calculator.aws> using the inputs below, and the shared link pasted into the placeholder.

Action required to finalize: open <https://calculator.aws>, recreate the line items in the *Inputs* table, click **Share**, and paste the generated link here:

AWS Pricing Calculator link: `PASTE_CALCULATOR_AWS_SHARE_LINK_HERE`

All prices below are **us-east-1, on-demand, USD**, and are indicative (AWS prices change). Two columns are shown: with the **AWS Free Tier** and **without** it.

Workload assumptions (baseline scenario)

Assumption	Value
Active hosts (monthly authenticated users)	50
Events per month	50
Guests per event	100
Total guests / RSVPs per month	~5,000
API requests per month (UI + RSVP + admin)	~100,000
Lambda invocations per month	~100,000
Avg Lambda duration / memory	200 ms @ 128 MB
DynamoDB reads / writes per month	~100,000 / ~100,000 (on-demand)
DynamoDB stored data	< 1 GB
S3 stored site	~5 MB
S3 GET requests per month	~500,000
SNS email notifications per month (invites + confirmations)	~10,000

Per-service estimate (monthly)

Service	Pricing basis (us-east-1)	Usage	With Free Tier	Without Free Tier
Lambda	\$0.20 / 1M requests + \$0.0000166667 / GB-s	100k req, 2,500 GB-s	\$0.00	~\$0.06
API Gateway (HTTP API)	\$1.00 / 1M requests	100k req	~\$0.00*	~\$0.10

Service	Pricing basis (us-east-1)	Usage	With Free Tier	Without Free Tier
DynamoDB (on-demand)	\$1.25 / 1M WRU, \$0.25 / 1M RRU + \$0.25 / GB	100k W, 100k R, <1 GB	~\$0.00*	~\$0.20
S3	\$0.023 / GB + \$0.0004 / 1k GET; 100 GB egress free	5 MB, 500k GET	~\$0.20	~\$0.20
Cognito	First 10,000 MAU free, then \$0.0055 / MAU	50 MAU	\$0.00	~\$0.00
SNS (email)	First 1,000 free, then \$2.00 / 100k + \$0.50 / 1M publishes	10k emails	~\$0.18	~\$0.18
Total	—	—	≈ \$0.40 / month	≈ \$0.74 / month

* API Gateway and DynamoDB free-tier allowances (1M HTTP API requests/month and 25 GB

- provisioned units, respectively) cover this baseline for the 12-month free tier.

Why it is so low

- Scales to zero** — no always-on compute; you pay per request/invoke.
- The dominant variable cost is **SNS email** and **API Gateway requests**, both linear in the number of guests/RSVPs.

Sensitivity (scale-up scenario)

If usage grows 20× (1,000 events/month, ~100,000 guests, ~2,000,000 API requests):

Service	Approx monthly (no free tier)
Lambda	~\$1.20
API Gateway	~\$2.00
DynamoDB	~\$4.00
S3	~\$2.00
Cognito (1,000 MAU)	~\$0.00 (under 10k free)
SNS email (~200k)	~\$4.00
Total	≈ \$13 / month

Notes / caveats

- Figures exclude taxes and any data-transfer beyond the 100 GB/month free egress.
- SES is **not** used (the lab account denies it); invitations use Cognito's built-in mailer for verification codes and SNS for invitations, which keeps email cost minimal.
- Re-run the calculator if region, memory size, or average duration change.